

1892-1916

MILESTONES

MASTER'S FIRST

Becomes the first woman and first African American to receive a master's degree in 1915.

BALL METHOD

Isolates chaulmoogra oil chemicals in 1916 to create an injectable form, which treats leprosy.

UNTIMELY DEATH

Dies in 1916, aged just 24, before her treatment is rolled out around the world.

WINS RECOGNITION

In 2000, 84 years after her death, the governor of Hawaii declares February 29 "Alice Ball Day."

US chemist Alice Ball created an injectable form of chaulmoogra oil extract. Known as "the Ball Method," it became the key treatment for the disfiguring disease leprosy until the 1940s.

Born in Seattle, Alice Augusta Ball studied chemistry at the University of Washington. She moved to Hawaii, and in 1915, became the first woman and the first African American to graduate from the College of Hawaii (now the University of Hawaii) with a master's degree in science. While teaching there, she researched a way to make injected chaulmoogra oil tolerable. Doctors knew the oil could treat leprosy, but when injected, it clumped under the skin, forming blisters. Ball exposed the oil's fatty acids to a catalyst to produce water-soluble ethyl esters (chemical compounds). When injected, the oil then dispersed. By 1922, her technique was widely used.

Ball died in 1916, a year after her discovery, and her college president was credited with the technique. In 2007, she was finally honored by the University of Hawaii with its Medal of Distinction.

"Who knows what other marvelous work she could have accomplished."

Paul Wermager, 2017

ALICE BALL

